

XINGJIAN (SCOTT) ZHANG

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Research engineer for efficient, reliable LLM inference: KV-cache reuse and compression, RAG, and rigorous evaluation.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

M.S. in Artificial Intelligence Engineering – Electrical & Computer Engineering

Aug 2026 – Dec 2027 (expected)

Coursework: Machine Learning for Engineers; Deep Learning for Computer Vision; Systems & Toolchains for AI Engineers

New York University

New York, NY

B.A. in Computer Science, Minors in Mathematics & Economics; GPA: 3.7/4.0

May 2026

INDEPENDENT RESEARCH (SOLE AUTHOR)

Reliable KV-Cache Reuse and Compression for LLM Inference

2026 – Present

- **SAFE-KV** (*AACL 2026 Findings*): diagnosed that non-prefix KV reuse in RAG keeps only 47.2%/37.8% of full-prefill answers (HotpotQA/2Wiki, Qwen2.5-7B); designed causal suffix repair restoring 99.8%/98.4%; shipped with 135 unit tests.
- **AuthorityKV**: built IH-CacheBench, showing H2O, Knorm and StreamingLLM (NVIDIA KVPress) reverse 2,055 of 7,200 instruction-hierarchy decisions at 20–40% KV; designed a training-free authority-tier constraint that cuts reversals to 0.
- **ProofKV**: built an evidence-to-KV compiler (typed proof selection, risk-calibrated routing, paged vLLM KV): +35.6 proof-complete recall on MuSiQue, +19.3 F1 over token-matched LLMingua-2, 1.16–1.84× end-to-end H100 speedup.

Grounded Retrieval-Augmented Generation and Evaluation

2026 – Present

- **EviDeLock**: built a training-free evidence channel for multi-agent RAG under retrieval conflict: +14.6 pts grounded accuracy over multi-agent debate (4 models, 4,800 examples) with 54% fewer tokens; +40.6 on held-out DRUID claims.
- **RAG decision audits**: ran a held-out audit of RAG system selection (31,200 generations, 6 retriever-reranker systems, 4 QA sets): risk-aware selection lifts winner retention 70%→82%; showed the LLM judge alone can flip the winner.

Faithful Chart Reasoning with Factor-Routed Experts

2026 – Present

- **FIRE-Graph**: factor-intervention programs + graph-composed MoE connector: factor-faithful accuracy 0.340→0.558; +3.5 pts on a 14k family-disjoint suite (McNemar $p = 7.3 \times 10^{-5}$); +0.8 on ChartQA-Pro (Qwen2.5-VL).

RESEARCH COLLABORATIONS

Efficient Sequence Modeling – with Prof. Eunsol Choi & Yilun Kuang (NYU)

May 2025 – Jan 2026

- Designed SubMLR, a multi-level low-rank attention architecture between quadratic attention and constant-memory linear RNNs; outperformed strong baselines on Zoology MQAR.
- Built hierarchical chunk-compression models on linear attention (~10% higher accuracy on synthetic tasks at matched FLOPs) and a FLOPs-matched training/evaluation pipeline on Slurm (PyTorch, Triton, flash-linear-attention).

LLM Reasoning Benchmark – remote collaboration with Prof. Haohan Wang (UIUC)

Feb 2025 – Jun 2025

- Co-built AdvGameBench (arXiv:2506.12012), a strategic-game benchmark of how LLMs plan and revise under resource budgets; defined process metrics (over-correction risk, correction success, over-budget rate) tied to win rate.

Chart Question Answering – remote collaboration with Prof. Qingyun Wang (William & Mary)

May 2025 – Jan 2026

- Extended ChartMoE (InternLM-XComposer2-VL-7B) with nine chart-type experts; built a self-verified Program-of-Thought ChartQA set for SFT and DPO/GRPO.

Transport Safety Modeling – remote collaboration with Dr. Faan Chen (Harvard)

Jan 2024 – Sep 2025

- Hybrid clustering + multi-criteria decision models for multi-country road-safety ranking: 4 journal papers (2 first-author); released 4 open-source, single-command reproduction packages.

PUBLICATIONS

- **X. Zhang**. SAFE-KV: Causal Suffix Repair for Behavior-Preserving Non-Prefix KV Cache Reuse in RAG. *AACL-IJCNLP 2026 Findings*.
- X. Yuan, **X. Zhang**, et al. Tracing LLM Reasoning Processes with Strategic Games. *arXiv:2506.12012*, 2025.
- Journals: **X. Zhang** et al., *Sci. Rep.* 15:36467 (2025); **X. Zhang** et al., *Results Eng.* 29:108543 (2026); Z. Guo, ..., **X. Zhang**, et al., *Knowl.-Based Syst.* 301:112286 (2024); J. Jin, ..., **X. Zhang**, et al., *Sci. Rep.* 15:35099 (2025).

SKILLS

LLM systems: PyTorch, HF Transformers, vLLM, KVPress, Triton, flash-linear-attention; LoRA, SFT, DPO/GRPO

Evaluation & statistics: benchmark construction, bootstrap/clustered CIs, McNemar & Holm tests, held-out calibration

Programming & infra: Python, C++, Java; Linux, Slurm/HPC, Git; A100/H100 GPUs **Languages:** Mandarin, English